

BANARAS HINDU UNIVERSITY

B.Com. (Hons.) — Semester III Examination, 2023-24

Paper No. BCH-216: Basic Statistics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. All questions carry equal marks (14 marks each).***Question 1.** “We compute ‘statistics’ from ‘statistics’ by ‘statistics.’” Explain clearly the three meanings of the word ‘statistics’ used in this statement and discuss its scope. [14]**OR**

Convert the following distribution into less than and more than cumulative frequency. Also draw its ogive:

Marks	0-5	5-10	10-15	15-20	20-25
Frequency	10	12	15	13	5

[14]

Question 2. Discuss the essential requisites of an ideal average. In the light of these requisites, state the merits and demerits of Mean, Median and Mode. [14]**OR**

- (i) The median of the following frequency table is 50. Find the missing frequencies f_1 and f_2 :

Class Interval	0-20	20-40	40-60	60-80	80-100	Total
Frequency	14	f_1	27	f_2	15	100

[9]

- (ii) The mean monthly salary paid to all employees in a certain company was Rs. 3,000. The mean monthly salaries paid to male and female employees were Rs. 3,200 and Rs. 2,200 respectively. Obtain the percentage of male and female employees in the company. [5]

Question 3. Explain the term ‘Dispersion’. Discuss its various measures. Also distinguish between absolute and relative measures of dispersion. [14]**OR**

- (i) From the prices of shares of A Company and B Company given below, state which is more stable in value:

A Company	55	54	52	53	56	58	52	50	51	49
B Company	108	107	105	105	106	107	104	103	104	101

[10]

- (ii) In a class of 10 students total of marks and variance are 400 and 20.25 while in another class of 40 students these values were 1200 and 68.89 respectively. Compare the marks of both groups. [4]
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Question 4. Explain computation method of Karl Pearson's coefficient of correlation with the help of imaginary figures. [14]

OR

Calculate coefficient of rank correlation from the following data and interpret it:

X	50	50	55	60	65	65	65	60	60	50
Y	11	13	14	16	16	15	15	14	13	13

[14]

Question 5. Differentiate between correlation and regression. Why are there two regression lines? Do they cut each other? If so, where? [14]

OR

From the following data calculate:

- (i) Two regression equations.
- (ii) Coefficient of correlation between X and Y.
- (iii) Most likely value of X when Y is 34.
- (iv) Most likely value of Y when X is 47.
- (v) Arithmetic mean of X and Y series.

X	48	50	53	49	51	55	53	49
Y	36	32	33	38	37	31	35	30

[14]